Reading Assignment 4
Report

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Quantifying Blockchain Extractable Value: How dark is the forest?

1 Initial Challenges

- Boundary of MEV: I was unsure which profits count as MEV (e.g., arbitrage, liquidation, sandwiching, reordering/insertion) and which are just normal market-making returns.
- Roles and incentives: The paper involves users, searchers, builders, validators/miners, and the protocol. I initially struggled to separate their incentives and conflicts clearly.
- Causal chain of extraction: The mechanism "transaction visibility opportunity detection ordering/priority control value extraction" was spread across sections and was hard to integrate at first.
- Realism of assumptions: I was uncertain how realistic the assumptions are regarding visibility, latency, rational behavior, and fee markets.
- Efficiency vs. fairness trade-off: Some MEV activity may improve pricing efficiency but may also hurt user welfare. I could not evaluate this trade-off confidently at first.
- Limits of mitigation methods: I was unclear whether proposed methods (private mempools, batch auctions, encrypted mempools, PBS-like designs) solve MEV or only reduce specific forms of it.

2 Process of Inquiry

My Prompt: “Explain the core definition of MEV in the context of this paper, and distinguish between: (a) normal market arbitrage (b) value extracted through ordering power (c) whether users are always harmed.”

My Analysis: The AI response was helpful because it separated categories by source of profit: If profit mainly comes from information processing and risk-taking without privileged ordering control, it is closer to ordinary arbitrage; If profit depends on control over inclusion/order/exclusion of transactions, it falls into MEV. This clarified my earlier misconception that “all arbitrage is MEV.” However, I also noticed one overgeneralization: the AI implied that MEV almost always harms users. A more precise view is that MEV often involves redistribution effects, and net welfare depends on context (e.g., some arbitrage can improve price alignment, while sandwiching clearly worsens execution quality for users)

My Prompt: “Break down one representative MEV attack (e.g., sandwiching) into a timeline: user tx enters mempool → searcher detects opportunity → front/back transactions are constructed → block is finalized. Also list the assumptions required at each step.”

My Analysis: The AI response was strong in process clarity. It turned an abstract mechanism into a concrete pipeline and highlighted key assumptions: User order flow is visible (public mempool or observable flow); Priority execution can be bought or negotiated (higher fee/tip, private channels, builder collaboration); The target transaction has enough price impact; Expected profit exceeds gas costs and failure risk.

My Prompt: “Among the mitigation approaches discussed in the paper, which ones reduce the creation of opportunities (prevention), and which ones mainly change value distribution (redistribution)? Compare them by target, cost, and residual risk.”

My Analysis: The response gave a useful comparison framework and helped me think beyond “works vs. doesn’t work”: Opportunity reduction: e.g., [placeholder: batch auctions / order-hiding designs], aimed at reducing predictable exploitable ordering. Value redistribution: e.g., [placeholder: PBS / auction-based ordering rights], aimed at making extraction more explicit and redistributing who captures it. Costs and residual risks: possible complexity, implementation overhead, centralization pressure, and new strategic behavior.

3 Synthesis and Lingering Questions

After iterative prompting and re-reading, my updated understanding of the paper is:
Root cause: MEV emerges from transaction ordering/inclusion control plus order-flow visibility.
Not one mechanism: MEV is a family of strategies (arbitrage, liquidation priority, sandwiching, reordering) shaped by market microstructure.
Core contribution of the paper (replace with exact wording from the paper): [placeholder: e.g., proposes model / quantifies extraction / compares mitigation regimes].
System-level tension: there is a three-way trade-off among efficiency, fairness, and decentralization.
Practical conclusion: most proposals are “harm reduction + redistribution,” not full elimination.